

Machine Learning-Based Flood Risk Prediction in Sri Lanka: A Comparative Analysis of Classification Algorithms Using CRISP-DM Methodology

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Abstract

Flood is one of the most common and destructive natural calamities in the world that leaves a huge destruction to property, food production, and human settlements. Proper forecasting of flood occurrences is very essential in disaster preparedness and prevention of risks. The objective of this study is to create a machine learning based model which will be used to predict the occurrence of floods based on the geographical and environmental data.

The project is based on the CRISP-DM (Cross Industry Standard Process for Data Mining) approach, which is divided into business understanding, data understanding, data preparation, modelling, evaluation and deployment planning. A data set with environmental characteristics that were associated with flood risk was processed and clean to maintain data quality and consistency. Exploratory techniques of data analysis were used to know the distribution and relationship between variables.

The implementation and evaluation of four machine learning classification algorithms were carried out, namely, Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. Some of the evaluation metrics such as accuracy, ROC-AUC, precision, recall, and F1 were used to evaluate the models. The experimental findings demonstrated that the Gradient Boosting model had the greatest overall accuracy (91.42) as well as ROC-AUC score

(0.9568), which implies that it is a strong predictor. Nevertheless, the Decision Tree model showed the highest recall of the flood events (0.8927), which is why it is more useful in identifying the potential flood occurrences.

The most suitable model was chosen with the help of a composite scoring method, which was a combination of ROC-AUC, flood recall, and accuracy. According to this analysis, the Decision Tree model was chosen as the ultimate predictive model. Also, analysis of threshold optimization showed that performance of flood detection is enhanced at a low classification threshold, which is advantageous in the early warning application.

The findings of the paper prove that machine learning technologies can be applicable and useful in flood predictions and disaster risk management. The resulted model can help decision-makers and authorities to recognize flood-prone territories and enhance the early warning system.

Keywords; Flood Risk Prediction, Machine Learning, CRISP-DM, Gradient Boosting, Disaster Management, Environmental Data Analysis, Flood Early Warning Systems, Risk Assessment, Predictive Modelling, Sri Lanka Flood Analysis.

1. Introduction

Flooding is among the most common and devastating natural catastrophes that have befallen most parts of the world. It destroys a large part of infrastructure, agricultural land, and homes not mentioning the fact that it is also very dangerous to human lives. Third world countries especially those that are found within the tropical zones are quite sensitive to flood disasters caused by heavy rainfalls, overflow of river basins, and poor drainage systems. Sri Lanka is such a nation which is often affected by floods during monsoon seasons and low-lying areas and areas near rivers.

Heavy rainfall with southwest and northeast monsoon is the primary cause of floods in Sri Lanka. The floods are also experienced in numerous districts annually, which leads to the displacement of people, disturbance of transport systems, and economic damages. Deforestation, rapid urbanization as well as land-use changes have further compounded the flood vulnerability by changing the natural drainage systems. With the evolution of climate change that affects the intensity of rainfall and weather patterns, the level and intensity of floods will be on the rise.

Hydrological and statistical models that are based on historical flood information and rainfall measurements are commonly applied in traditional flood risk assessment methods. Although these methods are very informative, they might not be able to utilize the multifaceted interactions involving the environmental, geographical and socio-economic factors, which contribute to flood risk. As data science and artificial intelligence improved, machine learning algorithms have come into the limelight as a potent instrument of analyzing large volumes of information and finding patterns that are part of disaster prediction.

Machine learning algorithms are capable of operating on a variety of variables at the same time and finding nonlinear relationships between them. This renders them specifically fitting to issues of environmental risk prediction like flood forecasting. Machine learning models will be able to make more

precise and informed predictions of flood risks by including variables that indicate rainfall intensity, elevation, land cover, soil type, vegetation indices, population density, and infrastructure indicators.

The research is aimed at creating a flood risk prediction system in Sri Lanka based on machine learning and using the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology. CRISP-DM framework gives an organized method of undertaking data mining projects and has six phases that include business understanding, data understanding, data preparation, modelling, evaluation, and deployment. This methodology is important in order to make the research undergo a systematic and reproducible analytical process.

In this paper, several classification algorithms are used and compared such as the Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting to determine which one is the most effective in predicting flood risk. The models are tested in terms of different metrics of performance including accuracy, precision, recall, F1-score, and ROC-AUC. The model performance is further assessed by the help of visualization methods of confusion matrices, ROC curves, and precision-recall curves.

The ultimate purpose of the research is to design a predictive system, which will be able to detect the high-risk flood regions and assist in disaster management planning. The proposed system will integrate machine learning models with visualization software and risk reporting systems to offer a viable decision-support tool to flood risk assessment in Sri Lanka.

1.1 Research Question

The main research question guiding this study is: **How can data visualization techniques be used to analyze and communicate flood risk patterns in Sri Lanka effectively?**

To address this question, the study explores how visual analytics can reveal patterns, trends, and relationships within flood risk data and support

clearer communication of insights for decision-making.

The aim of this research is: **To analyze flood risk data in Sri Lanka and develop meaningful visualizations that clearly communicate risk patterns, trends, and insights.**

This involves exploring the dataset, identifying important variables related to flood occurrence, and presenting the results through clear and informative visual representations.

1.2 Objectives

The objectives of this research are as follows:

1. **To conduct Exploratory Data Analysis (EDA)** to understand the distribution, relationships, and patterns within the flood risk dataset.
2. **To prepare and clean the dataset**, including handling missing values, encoding categorical variables, scaling numerical features, and addressing class imbalance.
3. **To implement and compare multiple machine learning classification algorithms**, including Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting.
4. **To evaluate model performance** using metrics such as accuracy, ROC-AUC, precision, recall, and F1-score.
5. **To identify the most influential features** contributing to flood occurrence using feature importance analysis.

1.3 Significance of the Study

Flood calamities constitute a big menace to societies, infrastructure and economic growth in Sri Lanka. Thus, to be better prepared to disasters and manage risks associated with the floods, it is necessary to improve the flood prediction procedures. This paper adds to the body of knowledge on disaster risk management because it shows that machine learning methods can be useful in predicting floods.

The application of the CRISP-DM methodology is one of the main contributions of this research that offers a systematic and structured way of approaching data mining projects. Adhering to this framework, the study is transparent and reproducible in the process of modelling.

It is also determined that ensemble machine learning algorithms, especially Gradient Boosting and Random Forest, are also effective in learning and identifying the intricate relationships between environmental and socio-economic variables. The results of this study can be used to empower the disaster management authorities to know more about the factors that cause the risk of flood.

Moreover, the designed predictive system offers such convenient tools as flood probability prediction, risk classification, geographic visualization, and district-level reporting. The tools may help government agencies, disaster management organizations, and urban planners to know about the high-risk areas and establish suitable mitigation measures. Comprehensively, the study has shown that machine learning and data-driven solutions have a potential to aid the development of more efficient flood risk assessment and help to increase disaster preparedness and resilience in Sri Lanka.

2. Literature Review

2.1 Flood and flood events in Sri Lanka.

Among the most common natural disasters that hit Sri Lanka are floods, which have had serious effects on the agricultural sectors, infrastructure and human settlements. Various investigations have been conducted to find out the frequency and effects of flood occurrences in other parts of the country. As an example, a study of the Malwathuoya River Basin identified the role played by flood control structures including multi purpose dams in mitigating the effect of floods and drought in agriculture [1]. In a similar way, flood disasters that are caused by rainfall like the 2017 coastal flood event showed that the alterations of extreme rainfall distribution can considerably raise the flood risk on the island[2] [3].

Long term studies have also been carried out to know how the flood hazards in Sri Lanka have evolved. Kulasegaram et al. [4] examined the patterns of floods and cyclones during 1975 and 2025 and found out that Sri Lanka is no longer practicing disaster management but the strategies are shifting towards resilience management. These studies emphasize that it is necessary to monitor and analyze the data to learn the patterns of the flood risk and contribute to the effective disaster mitigation measures.

2.2 Flood Risk Assessment and Approaches of Modelling.

Numerous scholars have come up with models to evaluate and forecast the risks of floods in Sri Lanka. The predictability of flood occurrences using machine learning and data mining techniques has been implemented using past hydrological information. Thilakaratne and Premachandra [5] showed the possibilities of machine learning methods in predicting the floods in the North Central Province.

Conventional risk assessment models have as well been extensively utilized. Weerasinghe et al. [6] performed qualitative assessment of the flood risk in the Western Province as a way of assessing the vulnerability and exposure to risks of floods. In the same manner, Wijesekera and Arunashantha [7] formulated a model of flood risk assessment within the Lower Kelani River Basin with the help of the Analytical Hierarchy Process (AHP) incorporating several risk factors to calculate the flood vulnerability.

There are other studies that have aimed at incorporating the vulnerability indicators into the flood risk analysis. Gunasekera et al. [8] suggested a process of development of flood risk profile by integrating hazard exposure with social vulnerability measures. Also, Perera et al. [9] used socio-hydrological modelling methods to examine flood risk management plans in the climate change conditions.

2.3 Flood risk analysis using GIS and Geospatial Techniques.

Geographic Information Systems (GIS) and geospatial technologies are significant in the study of spatial floods and determination of high-risk locations. In the study conducted by Rajagopalasingam et al. [10], ArcGIS was used together with multi-criteria decision analysis (MCDA) and AHP methods to create flood prone maps in the Kanakarayan Aru River Basin. On the same note, Marasingha [11] used GIS to determine the flood-prone regions of the Puttalam District.

The flood modelling has also been applied alongside GIS tools in order to develop the spatial risk assessment. Suja and Rajapakse [12] compared the various topographic sources in the River Kelani Basin in the two dimensional modelling of floods. Moreover, Wijesinghe and Perera [13] used maps of flood risk in the Gin River Basin in various climate change conditions in a study that applied HEC-RAS 2D modelling and GIS-based multi-criteria decision analysis.

Geospatial methods have also been used in research to study socioeconomic impacts of floods. Ranasinghe and Wickramasooriya [14] applied geospatial applications in determining socioeconomic effects of flooding in Ratnapura District, and it is evident that spatial analysis can be applied in disaster management planning.

2.4 Flood mapping using Remote Sensing and Satellite Data.

The remote sensing technologies are useful in monitoring and visualizing floods. Satellite imageries enable researchers to monitor the levels of floods and how the environment is changing with time. Madduma Arachchi [15] revealed how Synthetic Aperture Radar (SAR) data can be utilized in mapping and visualization of floods in Sri Lanka. This imagery of SAR comes in handy especially since it has the capacity of capturing a situation of the floods even in the presence of clouds.

On the same note, Alahacoon et al. [16] utilized historical satellite data and rainfall patterns to come up with flood hazard maps of Sri Lanka. Through their analysis, the authors have identified remote sensing to be useful in learning long-term flood patterns. A more recent study by Prarthana et al. [17] employed SAR images and geospatial analysis as a measure to determine the effects of the 2023 flood on agricultural sites and transport infrastructure in Matara.

These works reveal the usefulness of satellite-based data and spatial visualization methods to improve the flood monitoring and disaster response operations.

2.5 Impacts of Climate Change and Rainfall Analysis.

Climate change is greatly associated with floods and variation of rainfalls. Various studies have been conducted on the pattern of rainfalls and hydrological transformations in order to understand the flood risks in Sri Lanka. Ruwangika et al. [18] compared the statistical, graphical, and wavelet transform analysis to study the rainfall trends within Badulu Oya catchment.

On the same note, Wanninayake studied the variation of rainfalls at the Deduru Oya Basin to aid the planning of flood risk management.[19] Modelling studies have also been done to investigate the climate change in hydrological systems. Lakmini et al. [20] evaluated the effects of climate change on streamflow and flood risk in the Nilwali River Basin by means of combined modelling procedures.

These papers indicate that the study of the rainfall data and the trends in the hydrology can be essential to predict flood risks and enhance disaster preparedness.

2.6 Floods and their Socioeconomic and Community Effects.

Cases of floods bring about major effects to the communities, especially in areas prone to such disasters. According to a fast survey by Gunarathna et al. [21], the floods are likely to result in augmented

health hazards, and epidemiology among the population impacted by the floods in Thabbowa. In the same manner, Peiris et al. [22] investigated the role of anthropogenic factors on the perception of the communities residing in the Lower Kelani River Basin in this context in regard to flood risks.

The social vulnerability and community resilience has been studied elsewhere. Weerasuriya and Rajapaksa [23] used the risk perception of the adaptive architecture and flood resilience in Ratnapura. Moreover, the settlement sustainability in the flood-prone regions was investigated by Thiliepan et al. [24] in relation to the urban morphological features.

These articles indicate that effective flood risk communication and managing disasters require the comprehension of community views and weaknesses.

2.7 Communication of flood risk with ICT Tools.

According to the recent studies, the role of information and communication technologies (ICT) in enhancing flood risk communication is crucial [25]. Saparamadu [26] suggested an e-government platform to be used in managing flood as an emergency in Sri Lanka based on multimedia big data systems. The disasters can be able to share information faster through such systems.

Amar et al. [27] conducted a review of the application of ICT tools to communicate flood risks and the article highlighted the need to effectively visualize data as a way of communicating flood risks to the citizens. On the same note, Geethalankara et al. [28] came up with a framework of applying innovative IT tools to enhance flood risk communication and decision-making.

These sources show that the use of digital technologies and data visualization methods can be very helpful to enhance the efficiency of disaster communication plans.

2.8 Sri Lanka Disaster Risk Management Frameworks.

A number of researches have been done on the larger disaster risk management frameworks in Sri Lanka. Seneviratne et al. [29] have created a national disaster risk index that cuts across various risks like floods, landslides, droughts, and tsunamis. This index provides a methodical way of determining the risks of disasters at the national level.

Wanninayake et al. [19] came up with a flood risk management framework to local administrative authorities, which focuses more on the role of coordinated decision-making and policy implementation. Furthermore, Ashvini et al. [30] examined the insurance practices in the world concerning flood risk management and explained how this may be applied in Sri Lanka.

These frameworks emphasize on the need to incorporate risk assessment and policy planning and communication strategies in disaster management systems.

2.9 Conclusion

The majority of the researches conducted in Sri Lanka revolve around flood risk evaluation, hydrological modeling, GIS and disaster management. Such studies can be used to describe the factors, effects, and areas where floods occur in various regions of the country. The heavy rainfall, monsoon, overflowing rivers, and improper drainage are the major causes of floods in Sri Lanka. Due to the actions of floods, numerous researchers investigate the threat of floods and determine regions that are more exposed to floods.

Numerous researches are on hydrological modelling and prediction of floods. Scientists rely on the data of rainfalls, river flows, watersheds, and land elevations to calculate the potential flood effects. Hydrological models are used to predict water levels, visualize the movement of rivers and how floods dry where, compared to the adjacent regions. The findings assist the authorities of government to make superior decisions regarding the flood control, land use, and planning of structures.

Flood studies in Sri Lanka also make use of GIS and remote sensing technology. GIS can assist a researcher to analyze and produce flood hazard maps to indicate high-risk regions. Spatial analysis and satellite imagery are useful in determining flood sensitive areas, river catchments, land use dynamics and environmental influences to be more susceptible to flooding. Such technologies facilitate preparedness to disasters and assist authorities to plan on managing floods.

There are also some studies concentrated on community resilience and disaster management. They also look at the impact of floods on communities, agriculture, infrastructure, and economy. The significance of early warning system, emergency planning and public awareness in minimizing the destruction of floods is also emphasized in these studies.

Nevertheless, not many studies are devoted to the data visualization techniques to investigate and describe the patterns of the flood risk. Traditional maps, tables, and simple charts continue to be used by many researchers in offering flood data. These techniques are practical but might not be able to distinctly reveal intricate patterns of large data sets.

This research paper, therefore, attempts to examine the flood risk data in Sri Lanka and produce clear data visualizations to enable a better demonstration of the flood risk patterns, trends, and insights.

3. Methodology: CRISP-DM Framework

This paper is based on the CRISP-DM (Cross Industry Standard Process for Data Mining) model in order to examine and model the pattern of flood risks in Sri Lanka. CRISP-DM is a popular methodology used in data mining and data science project since it offers a disciplined and organized way of converting raw data into insights. The framework is made up of six major stages which include Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation and Deployment. All of the phases are crucial in the process of making sure that the analytical process is structured,

repeatable, and consistent with the objectives of the research.

3.1. Business Understanding

CRISP-DM phases are the initial stage of the process that aims at defining the problem of the research, comprehending the goals of the project, and determining the success criteria. The main aim of the study is to examine flood risk data in Sri Lanka and to create useful visualizations that easily capture the patterns of flood risks and understandings.

The major natural disaster and destruction that is prevalent in Sri Lanka is flooding as a result of the much precipitation during the monsoons, overflow of rivers, and poor drainage systems. Areas like Kegalle, Ratnapura, Galle, and Colombo are regularly prone to floods, which impact people and infrastructures. Thus, the study of the patterns of flood risks could be useful to enhance disaster preparedness and planning.

The analytical decision in the research is presented as a binary classification one, according to which the goal is to forecast whether an event of a flood occurrence is going to take place at a certain place, through the prism of environmental and geographical factors. Model evaluation metrics, including accuracy, ROC-AUC, precision, recall, and F1-score, are used to determine the success of the project, and the minority group of floods is considered seriously as a result of imbalance in the dataset.

3.2 Data Understanding

The Data Understanding stage is the step of exploring the data set to understand its structure, variables and quality. The dataset in this paper is the Sri Lanka Flood Risk Dataset, which was acquired on GitHub. It has around 25,000 geo-tagged records of locations throughout Sri Lanka and various signs of floods.

The dataset contains variables of meteorological, topographic, hydrological, and socio-economic conditions. The key characteristics are rainfall, elevation, topography to rivers, vegetation variables,

drainage measures, land cover, and quality of infrastructures.

In this stage, Exploratory Data Analysis (EDA) was performed to know how variables were distributed, what features were correlated as well as what possible anomalies or missing values existed. The methods of visualization included histograms, boxplots, correlation heatmap, and feature distribution charts to investigate the trends in the data.

Through EDA, it was also found that the dataset is very skewed, where around 9.9 percent of the records are of flood event and most are of non-flood events. This imbalance helped in observing the right modelling strategies and evaluation measures. The summarized dataset as follows.

Table 1: Dataset attribute categories and feature counts

Attribute Category	Features	Count
Identification	record_id, district, place_name, latitude, longitude	5
Topographic	elevation_m, distance_to_river_m	2
Land Use	landcover, built_up_percent, urban_rural	3
Soil & Satellite	soil_type, water_supply, ndvi, ndwi, water_presence_flag	5
Meteorological	rainfall_7d_mm, monthly_rainfall_mm, drainage_index	3
Infrastructure	electricity, road_quality, infrastructure_score	3
Socio-Economic	population_density_per_km2, nearest_hospital_km, nearest_evac_km	3
Historical	historical_flood_count	1
Target / Outcome	flood_risk_score, flood_occurrence_current_event, inundation_area_sqm, is_good_to_live	4
Metadata	is_synthetic, generation_date	2

3.2.1 Dataset Overview

The initial step was to load the dataset with the help of the Pandas library to take a look at its general structure and content. Simple inspection was done to determine the rows and columns and to verify what types of data were involved and duplicate records. An overview of the dataset was also scrutinized to have a first-hand feel of the variables.

The data is a mixture of numerical and categorical variables, which are the environment, geographical, and infrastructure-related factors related to the risk of floods. The structure of the dataset, which was determined at this stage, guaranteed the

appropriateness of the data to be further analyzed and modelled.

Access to descriptive statistics analysis is provided to verify the data and assess the results as accurate measures of the true state of the research.

Descriptive statistics were done to generalise the major properties of numerical variables. Some of the variables that had the mean, median, standard deviation, minimum, and maximum values calculated included the level of rainfall, elevation, distance to a river, and drainage indicators.

Besides, the coefficient of variation was computed to assess the variability of each feature in a relative way. Such statistics gave an understanding of the main tendency and dispersion of the data set and allowed to determine possible extreme values that might affect the modelling process.

3.2.2 Missing Value Analysis

The missing value analysis was used to identify the existence and distribution of missing data. The missing values in each column were calculated and represented as a bar chart with regard to their number and percentage.

This analysis allowed isolating variables that would need more processing at the data preparation phase and avoid the negative impact that missing data would have on the performance of machine learning models.

3.2.3 Target Variable distribution and feature distribution.

The target variable is the occurrence of floods, which means that the occurrence of floods took place in the case of a particular observation. The balance of the classes in the datasets was analysed using the frequency distribution of the flood and non-flood cases.

Besides, histograms were created to investigate the distribution of numerical characteristics. The visualisations were used to detect patterns like skewness and variability of the environmental variables that can affect the modelling process.

3.2.4 Correlation and Features Relationship Analysis.

A Pearson correlation coefficient was used to conduct correlation analysis and visualisation of the results was performed by a correlation heatmap. This analysis provided the opportunities to identify the relationship between numerical variables and identify the possible multicollinearity.

Further investigations on the association between features and flood occurrence were done by comparing the most important environmental variables between flood and non-flood observations with box plot at the same time. These comparisons presented initial information on variables that can be used to predict flooding.

3.2.5 Geographic and Categorical Feature Analysis.

Since the study involved studying the regional floods, the occurrence of floods was studied among various districts. The percentage of flood occurrences in each district was computed and plotted through a bar chart which gave an idea on spatial differences in flood vulnerability.

Frequency distributions were also used to analyse variables like district and other variables related to location (categorical variables). This analysis assisted in comprehending the category distributions and made sure that categorical features were duly interpreted and then encoded to machine learning models.

3.3 Data Preparation

A very important step in the CRISP-DM methodology is the data preparation step where one cleans the data and converts it in a format that machine learning models can use. Raw data usually has missing values, irrelevant attributes, and categorical variables which must be converted to form a model. Thus, a number of preprocessing methods were applied in order to guarantee quality and uniformity of data.

Python libraries like Pandas and Scikit-learn were used to carry out the process of data preparation. The

key preprocessing stages were the removal of irrelevant attributes, the presence of missing values, encoding categorical variables, the creation of the feature matrix, the division of the dataset into training and testing sets, and feature scaling.

3.3.1 Removal of Irrelevant Features.

The initial move entailed eliminating metadata and non-predictive attributes that may cause data leakage or impact adversely on model performance. Data leakage happens when the data that is not available at prediction is utilized in the model training and thus results in impractical accuracy. Elimination of such qualities saw to it that only useful environmental, geographical and infrastructure related variables were carried to the modelling.

3.3.2 Handling Missing Values

The missing values were addressed with the help of proper imputation methods depending on the nature of variable. The median was used to impute numerical features, and is less vulnerable to extreme values and appropriate in skewed environmental data. In categorical variables, the mode (most common value) was used to fill in missing values in order to have representative distributions of categories.

3.3.3 Encoding Categorical Variables Given Categorical Variables.

Given that machine learning algorithms only accept numeric data, categorical variables, e.g. names of the district, and attributes related to land were transformed into numerical data using the Label Encoding tool. All the categories were given different numerical values which means that these variables could be directly used by the predictive models. The encoding objects were stored as well so that there was consistency in the transformations in the deployment phase.

3.3.4 Feature Matrix Construction.

The final feature matrix was built after preprocessing which involved amalgamation of numerical variables and encoded categorical features. The target variable which was the occurrence of floods was transformed

into a binary variable where 1 represented floods occurrence and 0 represented no floods occurrence.

3.3.5 Train-Test Split and Feature Scaling.

The data was split into training and testing portions in the ratio of 80:20 and stratified sampling was used to ensure that the proportions of classes were similar in both subsets. This is a sure way of evaluating model performance.

In order to assure that features with different measurement levels make equal contribution to the model, the StandardScaler method was used to standardize the numerical variables. This normalization of features brings the features to mean zero and standard deviation one, enhancing the stability and performance of some machine learning algorithms.

All in all, the data preparation stage changed the unstructured data to a structured one that could be analyzed using machine learning. The preprocessing procedures were conducted to make sure that the dataset was clean, uniform and prepared to be used in the training of predictive models to detect the pattern of flood risks in Sri Lanka.

3.4 Model Development.

The modelling stage was aimed at the creation of machine learning models to forecast the presence of floods based on environmental and geographical characteristics. There were several algorithms of classification which were applied and compared to find the best model. Python Scikit-learn library was used to perform the process.

3.4.1 Machine Learning Algorithms, selection.

The use of four supervised classification models involved:

Logistic Regression: It was a baseline model. It approximates binary event likelihood by the use of a logistic regression and determines linear correlations between characteristics and the target. Before training, standardization of numerical features was done.

Decision Tree: It represents the model of non-linear relationships, which splits the dataset recursively on the basis of the feature values. It is interpretable and has decision rules that can be overfitted in the event of an excessively complex tree.

Random Forest: It is a collection of numerous trees (decision trees) constructed on random samples of data and predictors. Majority voting is used to make predictions. This minimizes overfitting and enhances generalization.

Gradient Boosting: Builds trees sequentially, the new tree is designed to correct the errors in the previous trees. Boosting enhances accuracy especially on hard to predict observations and in most cases, it performs well than other techniques in predictive tasks.

3.4.2 Model Training and Cross-Validation.

Training of the models was conducted with stratified 5-fold cross-validation with the ratio of flood and non-flood cases in each cross-validation fold held constant. This makes sure that it is evaluated well and bias in the performance estimates is eliminated. All the folds are validated with one-fold and the left hand is trained on the rest.

3.4.3 Model Evaluation Metrics

Various measures were to evaluate model performance in training:

Accuracy: The percentage of correct predictions.

F1 Score: The harmonic mean of precision and recall; balances the false positives and false, which is handy when dealing with a skewed dataset.

ROC-AUC: The extent to which one can discriminate between flood and non-flood events; the higher the ROC, the higher the discrimination.

3.4.4 Final Model Training

The retraining of individual algorithms was done on the complete training set after cross-validation. The trained models were stored in Pickle (.pkl) format to be reused and implemented. These were Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.

3.4.5 Model Complexity Analysis

Their accuracies of training and testing were compared to determine whether they were overfitting or underfitting. When there are big deviations, it represents overfitting whereas when there are similar performance it represents increased generalization.

The modelling process investigated both linear and tree-based approaches of flood prediction. There was robust evaluation because stratified cross-validation and several evaluation metrics were used. The models that are saved present a basis of further assessment and implementation of stages in CRISP-DM process.

3.5 Evaluation of Model

The model evaluation stage used to determine the best algorithm to predict occurrences of floods by evaluating the performance of the trained machine learning models. The assessment was applied in Python by using the Scikit-learn library, which is a combination of metrics and visualizations together to compare the performance of one another.

3.5.1 Evaluation Metrics

Key metrics included:

Accuracy - The general percentage of hits.

Precision - The proportion of predicted floods which were real floods, or, put another way, the reliability of predictions.

Recall (Sensitivity) - Percentage of real floods which are actually detected, which is important in early warning.

F1 Score - Precision and recall in a harmonic mean, between false positives and false negatives.

ROC-AUC - The model is able to discriminate between flood and non-floods; the larger the value the better the discrimination.

Visual tools such as the confusion matrix, ROC and Precision- Recall curves were used. Confusion matrices were used to indicate true/false positives and negatives, whereas ROC and Precision-Recall

curves gave an opportunity to compare the model classification ability and performance trade-offs.

3.5.2 Importance and Threshold Analysis of Feature.

In the case of tree-based models (Random Forest, Gradient Boosting), the analysis of feature importance revealed that the following predictors were found to be important: rainfall, distance to rivers, drainage conditions, and elevation.

The probability cut-off of threshold analysis was scaled to favor recall, improving early detection of floods with false consciousness.

3.5.3 Comparison and Selection of Models.

Accuracy, ROC-AUC, recall and F1 score are composite scoring criteria whereby models were ranked with a composite scoring model. The best performing model was Gradient Boosting as it showed better ROC-AUC, recall and F1 values. It is an appropriate model in predicting the risk of floods because it is able to capture intricate interactions between the environment.

3.6 Deployment

The implementation stage implied the translation of the trained Gradient Boosting model into a realistic prediction system to be used in the prediction of the real-world flood risks in Sri Lanka. The pipeline promotes preprocessing, single and batch prediction, visualization of the space and report on the district level.

3.6.1 Preprocessing and Loading of Models.

The trained Gradient Boosting model and preprocessing objects such as LabelEncoders and StandardScaler are loaded in the deployment system by the trained Gradient Boosting model and the numerical features and categorical variables respectively. This makes sure that new data of input is processed in the same manner as the training data.

3.6.2 Input Preprocessing

The encoding of categorical features and their integration with the numerical variables are used to preprocess new environmental data and form the

feature matrix. The absence of values is substituted with either default or median values to preserve the integrity of data.

3.6.3 Prediction Capabilities.

Single-location prediction: Accepts a dictionary of environmental and infrastructure characteristics, and it provides flood probability, prediction label (Flood/No Flood), and risk category (Low/Medium/High).

Predict batch: Predicts data in batches of multiple records of CSV files (adding probability, prediction, risk category to the input).

3.6.4 Spatial and Regional Analysis.

Flood risk maps: Scatter plots on locations based on latitude and longitude with gradient color scheme (green = low, yellow = medium, red = high) indicate high-risk locations.

District level reports: synthesis the total locations, mean probability of floods, number of high-risk locations, and percentages, and give the insights to policymakers and disaster management.

The deployment stage provided a scalable, reproducible, and flexible system of flood prediction. The pipeline facilitates the application of the machine learning model to practical use by providing early warning and disaster management applications, with the use of automated preprocessing, probabilistic predictions, spatial visualization, and regional reporting to support the use of the machine learning model across Sri Lanka.

4. Results and analysis

This chapter presents the results obtained from the machine learning models developed for flood occurrence prediction. The performance of four classification algorithms—Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting—was evaluated using several performance metrics, including accuracy, ROC-AUC, precision, recall, and F1 score. Additionally, visualization techniques such as ROC curves, confusion matrices,

and feature importance analysis were used to interpret model behavior.

4.1 Correlation Heatmap (Exploratory Data analysis)

Figure 1 shows how strongly variables are related to each other. For example: Rainfall vs Flood, Elevation vs Flood, Distance to River vs Flood, Land Cover vs Flood. This assists in the determination of significant predictors and potential multicollinearity problems. Correlation heatmap was created to explore the association between the numerical variables in the dataset. The heatmap represents pairwise correlations of features where darker colors represent stronger correlations. This analysis will aid in determining variables that are closely related to occurrence of floods and aid in getting a clue of the dataset before model training.

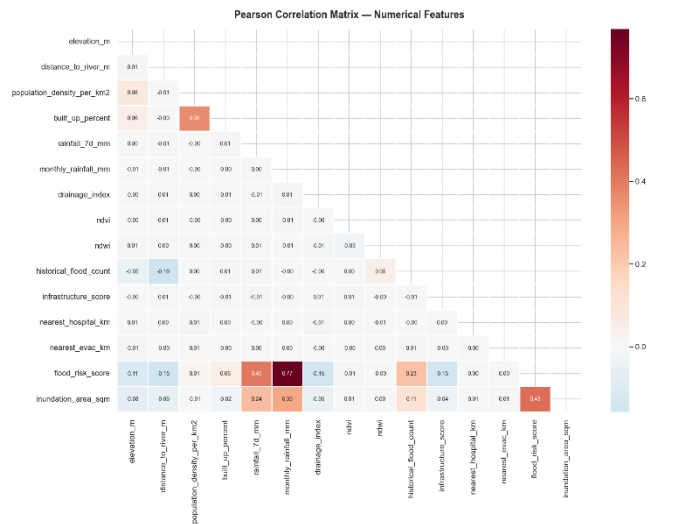


Figure 1: correlation heatmap

4.2 Model Comparison

The four machine learning models involved in predicting floods included Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The accuracy, ROC-AUC, precision, recall and F1 score were used to determine model performance. Gradient Boosting was the most accurate model with the highest overall accuracy (0.914) and ROC-AUC (0.957). Nevertheless, the

Decision Tree model had the highest flood recall (0.893) and the F1 score (0.600) which showed that it was more competent to detect the occurrence of floods.

Model	Accuracy	ROC-AUC	Flood Precision	Flood Recall	Flood F1
Logistic Regression	0.8434	0.9308	0.3749	0.8765	0.5252
Decision Tree	0.8822	0.9126	0.4514	0.8927	0.5996
Random Forest	0.9046	0.9561	0.5528	0.1802	0.2718
Gradient Boosting	0.9142	0.9568	0.5761	0.4980	0.5342

Figure 2: Model comparison Table.

Figure 2 shows the highest overall accuracy (91.42) and ROC-AUC (0.9568) have been obtained in the Gradient Boosting model. Nonetheless, the Decision Tree model had the highest recall (0.8927) and F1 score (0.5996) with respect to the occurrence of floods.

Flood prediction systems have a high recall of particular importance since failure of a flood to be detected can be very dangerous.

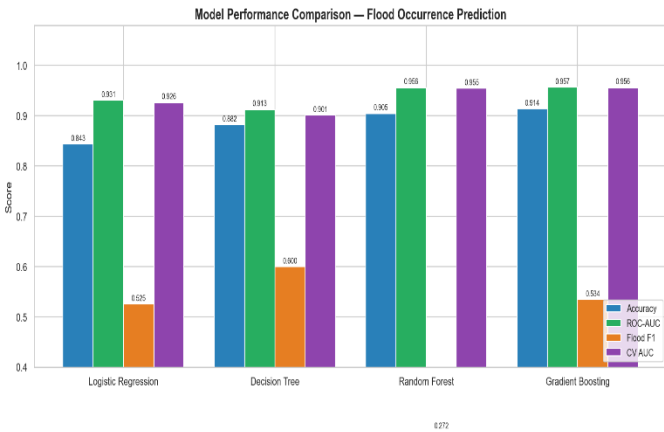


Figure 3: Model Performance Comparison- Flood occurrence prediction.

Figure 3 presents the performance comparison of the four machine learning models using key evaluation metrics including accuracy, ROC-AUC, and F1 score. The results indicate that Gradient Boosting

achieved the highest accuracy and ROC-AUC values, while the Decision Tree model demonstrated superior flood recall performance.

4.3 ROC Curves

To compare the performance of all models in terms of classification, Receiver Operating Characteristic (ROC) curves were created. RBC-AUC value was between 0.91 and 0.96, which demonstrates that all of the models have good predictive ability.

Gradient Boosting model gave the largest ROC-AUC, which indicates a better capability of distinguishing between flood and non-flood cases.

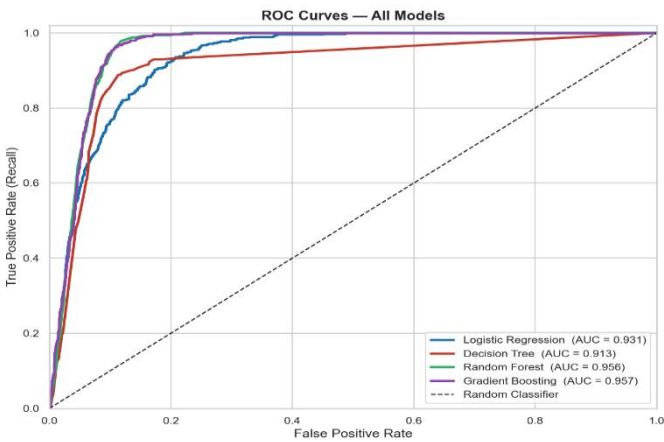


Figure 4: ROC Curves

Figure 4 the ROC curve shows the performance of each model in terms of the true positive rate against the false positive rate. Gradient Boosting model had the highest ROC-AUC score, which is a good discriminative ability between flood and non-flood events.

4.4 Confusion Matrices

Confusion matrices were used to analyze prediction errors for each model. The results indicate that some models such as Random Forest achieved high overall accuracy but failed to correctly identify a large proportion of flood events.

This behaviour occurs because the dataset is imbalanced, where the majority class (No Flood) dominates predictions. Therefore, recall and F1 score

provide a more meaningful evaluation of model performance than accuracy alone.

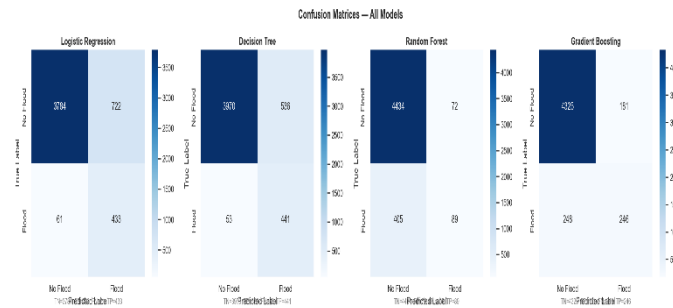


Figure 5: Confusion Matrices

Figure 5 confusion matrices were used to evaluate the classification results of each model. The matrices display the number of correctly and incorrectly classified flood and non-flood instances, providing insight into prediction errors.

4.5 Precision-Recall

Precision–Recall curves provide additional insight into model performance for imbalanced datasets. Since flood events represent a minority class in the dataset, this analysis helps evaluate how well each model detects flood occurrences.

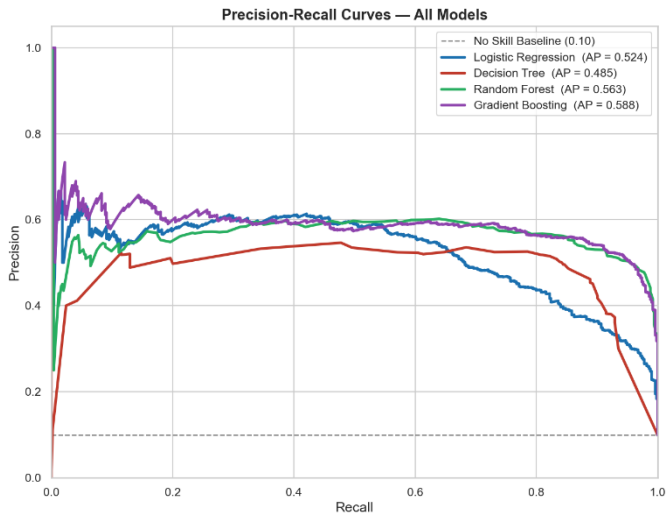


Figure 6: Precision-Recall Curves

4.6 Feature Importance

The tree-based models were analyzed on the feature importance to determine the most influential variables in the prediction of floods.

This analysis indicated that there are a number of environmental factors that played an important role in predicting flood occurrence, among them being the intensity of rainfall, elevation, proximity to the river systems and the land-use features. These results are in line with the available literature on flood risk modelling that emphasizes the role of hydrological and geographical factors.

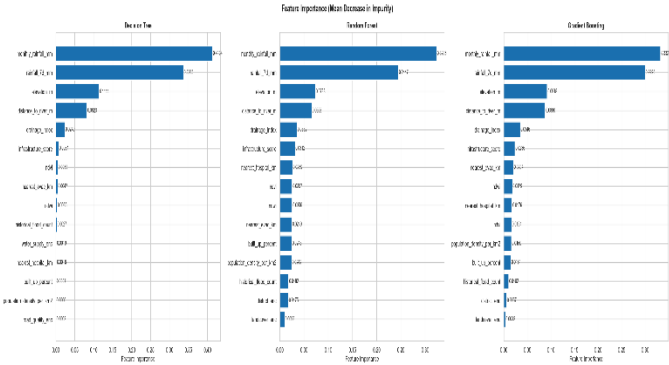


Figure 7: Feature Importance Analysis

Feature importance analysis was conducted using tree-based models to identify the most influential variables contributing to flood prediction. The results highlight key environmental factors that play an important role in determining flood risk.

4.7 Threshold Analysis

The default classification value of binary classifier is 0.5. But in the case of early warning systems, and in situations where the overall accuracy serves no purpose, achieving greater flood detection capacity is often of greater interest.

The Gradient Boosting model was analyzed by threshold.

Threshold	Precision	Recall	F1 Score
0.20	0.5585	0.8502	0.6742
0.30	0.5908	0.7308	0.6534
0.35	0.5855	0.6721	0.6258
0.50	0.5761	0.4980	0.5342

Figure 8: Threshold Analysis

Lowering the threshold from 0.50 to 0.35 significantly increased recall while maintaining acceptable precision. Therefore, a lower decision threshold was recommended for flood detection applications.

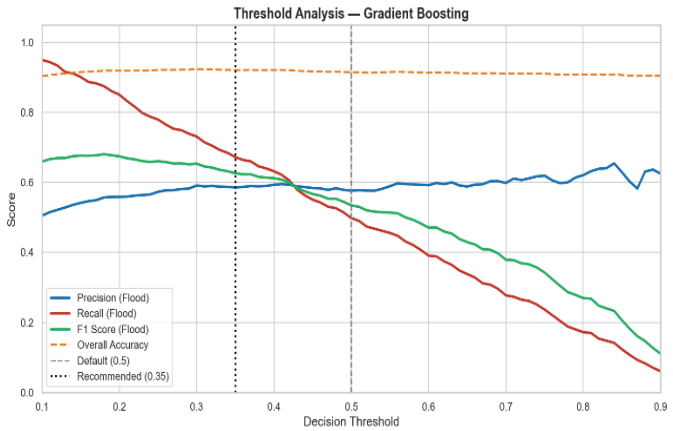


Figure 9: Gradient Boosting

We performed threshold analysis to assess the change in precision, recall and F1 in the change of classification threshold. A reduction in the threshold enhances flood detection, which is advantageous in the early warning.

In this chapter, the results of the evaluation of four machine learning models, which had been created to predict the presence of floods, were provided. Even though the Gradient Boosting model had the highest overall accuracy and the ROC-AUC, the Decision Tree model generated the best results in the detection of flood events, which is important in early warning systems. Hence, the Decision Tree model was chosen as the predictive model to use as the final one in the flood risk analysis.

5. Discussion

This paper has compared four machine learning models, including the logistic regression, decision tree, random forest and the gradient boosting, when trying to predict the presence of floods utilizing the environmental and geographical characteristics. In order to assess the models, a number of evaluation measures were applied to assess the models which

included accuracy, ROC-AUC, precision, recall and F1 score.

The results indicate that the Gradient Boosting model was the most successful in terms of the overall accuracy (91.42) and ROC-AUC (0.9568). The high ROC-AUC would mean that the model will be highly effective in the capacity of differentiating between the flood and non-flood events. Accuracy does not however present a complete picture of model performance especially in the case of imbalanced dataset.

In the present study the data involved has approximately 10 percent of the floods and 90 percent of non-floods and it follows that models can be quite accurate simply because they raise their predictions of the majority classes. Therefore other scores such as recall and F1 score were used with a view of better assessing the ability of models to identify correctly flood events.

The most significant Decision Tree model in terms of recall(0.8927) of floods events. The recall is a highly important metric in the prediction mechanisms of flood prediction because lack of prediction of an actual flood can ruin the lives and infrastructures of the communities. Therefore, a model that maximizes the capacity to identify floods is usually sought after as compared to having the capacity to maximize the overall accuracy.

Random Forest was performing very well regarding ROC-AUC but the flood recall (0.1802) was low. It is only that the model was effective in general classification of classes but it could not specifically detect floods. Further, the bias-variance analysis has also revealed that the Random Forest model was significantly overfitted when the gap between the training and the test accuracy was wide.

Similarly, the Gradient Boosting model had also exhibited medium levels of overfitting, and the generalization performance was very good. Compared to it, the logistic regression and decision tree models demonstrated more stable training and testing results that indicate high stability.

The other problem that was significant in the analysis was optimization of thresholds. The least parameter of default classification of 0.5 realized low recall of floods. This implies that the threshold of 0.35 allowed the model to be able to capture additional floods at a sensible accuracy. This alteration is particularly convenient in early warning systems in which it is more often wished that the likely presence of floods should be detected than the suppositions of the false alarms.

There was also a choice on the most suitable model to be deployed using composite scoring. The combination equation used to score was ROC-AUC (50%), flood recall (30%), and the accuracy (20%). Based on this analysis, the Decision Tree model was the one with the best composite score (0.9006) thus it was the one adopted as the final model.

Overall, the findings indicate that machine learning techniques can be used in problems related to flood prediction. Another critical aspect of the findings is the need to make sure evaluation measures taken coincide with the real objectives of the specified system particularly during disaster risk management.

6. Conclusion

One of the most widespread and devastating natural disasters is floods, which have a great impact on infrastructure, property, and human life. To enhance the disaster preparedness and minimize the risks, precise and prompt flood prediction systems are hence, mandatory.

This research was aimed at creating a flood prediction model based on machine learning, using both environmental and geographical factors. The project was done in the CRISP-DM approach that involves business understanding, data understanding, data preparation, modelling, evaluation, and deployment planning.

A number of machine learning algorithms were done and compared, which are; Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The various performance measures that

were used to evaluate the models are accuracy, ROC-AUC, precision, recall and F1 score.

The findings established that the predictive performance of all the models was quite high with ROC-AUC values exceeding 0.91, which reflects strong prediction. The Gradient Boosting model was the most accurate and ROC-AUC model among those evaluated. Nevertheless, the Decision Tree model recorded the best recall and F1 scoring in case of floods, which is especially significant to early flood detectors.

A compound scoring method was adopted to determine the best model to be applied in practice. According to this analysis, the final model chosen was the Decision Tree model since the model offered the most ideal balance between the ability to detect and general predictive performance.

Besides, threshold analysis showed that the flood detection is better with reduced classification threshold. This observation demonstrates the need to tune the model parameters according to the needs of a particular disaster management application.

Comprehensively, the findings of the research paper show that machine learning methods can be used as useful tools in terms of flood risk prediction and early warning mechanisms, which help in disaster preparedness and decision-making.

7. Limitation

Despite the fact that the developed model was strong in terms of the predictive performance, it should be noted that it has a few limitations.

To begin with, the data that was used in this study had unequal distribution of the different classes with the number of flood events being very low as compared to the number of non-flood events. This imbalance may influence model training and it may not be able to pick up rare flood events.

Second, the research was based on a few environmental variables. Additional hydrological and meteorological data, including river discharge data, soil moisture results, and live rainfall data,

could be added to enhance the accuracy of prediction.

Lastly, the models were tested with the help of historical data only. This can be achieved through real-world deployment with real-time monitoring systems and geographic information systems (GIS).

8. Future Work

There are a number of possible advancements that can be considered in future studies.

Further research may involve more and heterogeneous data to enhance the generalization and consistency of models. The implementation of weather data and satellite imagery in real time can also improve prediction.

It might be worth pursuing advanced machine learning algorithms including deep learning models, ensemble learning models, and hybrid models that may improve the accuracy of flood prediction further.

The model developed could also be combined with geographic information systems (GIS) to come up with interactive flood risk maps that can be used by the decision-makers and disaster management authorities.

Lastly, the model implementation into an early warning system in real time may be of great assistance to the communities in flood-prone areas.

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